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Roll No

AU-704 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Combustion and Heat Transfer
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is Abnormal combustion?
b) What is air-swirl?
2. a) Explain the heat transfer, temperature distribution and thermal stress in piston of IC engines.
b) What are the variation of gas temperatures effect of engine load on piston temperature heat rejected to coolant quantity of water required.
3. a) Explain the various stages of combustion in CI engine.
b) Derive the expression of heat transfer through a piston crown.
4. a) With neat sketches explain the phenomenon of knocking in S.I engines.
b) Briefly explain the need, characteristics and types of cooling systems.
5. Write down basic laws of conduction convection and radiation.
6. Define Laminar and Turbulent flow.

OR

Discuss in brief the theory of engine heat transfer with supporting equations.

7. a) Write Fourier equation.
b) What is Thermal Diffusivity?
8. Write short notes on any two of the following:
 - a) Stefan Boltzmann law
 - b) Kirchhoff's law
 - c) Plank's law
 - d) Dielectric heating
